Chapter 31

Restoring Driving and Community Mobility

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LEARNING OBJECTIVES

This chapter will allow the reader to:

1. Conceptualize a definition of driving as an instrumental activity of daily living.
2. Understand the implications of motor vehicle crash statistics and trends over a decade in the United States.
3. Articulate the roles of the occupational therapist generalist and specialist in driver rehabilitation, specifically related to screening, assessment, and intervention.
4. Integrate and synthesize the knowledge on assessment and intervention via a case study example.
5. Understand the impact of community mobility on participation in essential and meaningful activities.
6. Identify the role of occupational therapist in addressing community mobility issues.
7. Understand assessment and intervention processes to improve the person–environment fit for community mobility.

CHAPTER OUTLINE

Driving as an Instrumental Activity of Daily Living
Overview of Crashes in the United States
Driver Rehabilitation
Interventions to Support Driving
Community Mobility
Case Study: Julio
Assessment of Community Mobility
Interventions to Support Community Mobility
Case Study: Tristen

TERMINOLOGY

Community mobility: moving around in the community and using public or private transportation, such as driving, walking, bicycling, or accessing and riding in buses, taxi cabs, or other transportation systems.[1]

Community participation: active involvement in activities that are intrinsically social and either occur outside the home or are part of a nondomestic role.[2]

Comprehensive driving evaluation: a complete evaluation of an individual's driving knowledge, skills, and abilities by a health care professional that includes (1) medical and driving history; (2) clinical assessment of sensory/perceptual, cognitive, or psychomotor functional abilities; (3) on-road assessment, as appropriate; (4) an outcome summary; and (5) recommendations for an inclusive mobility plan including transportation options.[3]

Driver rehabilitation specialist (DRS): professional who provides clinical driving and driving mobility equipment evaluations and intervention to develop or restore driving skills and abilities[1]; may or may not have a health professional background; with a health professional background, a DRS can provide the comprehensive driving evaluation.

Driver rehabilitation specialist, certified: a DRS certified by the Association for Driver Rehabilitation Specialists (ADED) to hold a specialty certification in driver rehabilitation.

Fitness to drive: a characteristic of a driver, defined by the absence of any functional (sensory/perceptual, cognitive, or psychomotor) deficit or medical condition that significantly impairs an individual's ability to fully control the vehicle while conforming to the rules of the road and obeying traffic laws and/or that significantly increases crash risk.[4]

Occupational therapist generalists: professional who provides clinical driving evaluations or applies specific assessment tools in order to assess fitness to drive as an instrumental activity of daily living and intervention to develop or restore driving abilities.[5]

Paratransit: a supplemental form of transportation required by the American with Disability Act (ADA) provided by public transit agencies for individuals with disabilities who cannot use fixed-route transit services because of a disability.[6]

Personal mobility devices: devices used for community mobility when a person cannot walk or has limited walking ability such as a manual or electric wheelchair.

the most Complex (IADL) لأنه يحتاج

Driving as an Instrumental Activity of Daily Living

Driving is an instrumental activity of daily living (IADL) that requires intact visual, visual attention, visual perceptual, cognitive, executive, other sensory, and motor functions; all executed in a coordinated manner within a complex, dynamic, and unpredictable environment, while achieving control over the vehicle to steer it cautiously and safely in the flow of traffic and while observing the rules of the road and traffic regulations. This task, therefore, represents an integration of the person, the vehicle, and the environment domains and, as such, is embedded in the scope of occupational therapy practice. Driving is considered a privilege, not a right, and is one of the few IADLs that can lead to death or serious injury. But the benefits of being able to drive are multiple. Mainly, driving is an occupation enabler and a mediator of autonomy, freedom, and independence. For older drivers, studies associate mobility afforded by driving with increased life satisfaction, quality of life (QoL), autonomy, and well-being.[7–9] In contrast, driving cessation is associated with poor health trajectories, including increased rates of depression, limited life-space mobility, early nursing home admissions, and premature death.[10–14]

ما نقدر to predict السيئة والاستثناء الي تطلع فجأة

حافظ على حياته وعلى الـ safety وعلى الناس تبقى السيارة والسواقة

تعطي استقلالية Freedom و

Our society is standing on the brink of a transportation revolution with the advent of the autonomous vehicles into our society. Although the highly autonomous vehicles may not be available for societal use for another decade or more,[15] we can anticipate that the task of driving, as well as the role of the driver, will change accordingly. Specifically, in the future, the task of driving will be executed by the autonomous vehicle, and the driver will, therefore, have to surrender personal control, show confidence in technology, and trust in the autonomous system. The role of the driver may change from a driver in a standard vehicle, to an operator in a semi-autonomous vehicle, or a passenger in a highly autonomous vehicle. Moreover, drivers and occupational therapists who may assess such drivers will have to be savvy and understand the lingo related to the evolving vehicle technology. Until such time, however, occupational therapists need to understand that driving has inherent risks and that they can make a difference in detecting at-risk drivers. To contextualize the scope of such risk, we will provide an overview of the crash statistics in the United States.

Overview of Crashes in the United States

According to the National Highway Traffic Safety Association (NHTSA), 35,092 road fatalities occurred in the United States in 2015, which is a 7.2% increase over 2014.[16] This is the largest percentage increase recorded in nearly 50 years with a fatality rate of 10.9 per 100,000 inhabitants. Provisional data from the first 9 months of 2016 indicate an additional 8% increase in traffic fatalities over the same period in 2015. Notwithstanding the loss of life, the number of injury crashes, and those who were seriously injured, also increased substantially. The NHTSA estimated that the cost associated with these road fatalities and injuries accounted for 242 billion dollars in 2010. When societal harm is factored in, then the cost increases to 836 billion dollars, which is about 6% of the gross domestic product in the United States.[16] The most vulnerable road users are the teens and the older drivers, as can be

seen from Figure 31-1. Since 2010, teens (18–20 years of age) had the highest fatality rates per 100,000 population, followed by older persons (65 years and older).

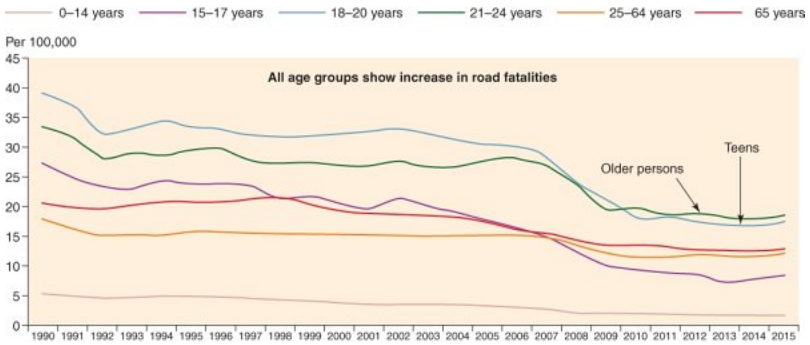

Figure 31-1 Road fatality rates by age group. Deaths per 100,000 inhabitants in a given age group, 1990 to 2015.74

Occupational therapists are positioned to help turn these statistics around though early detection and intervention. Specifically, human error accounts for 94% of crash-related fatalities or injuries.[16] Human errors, which may include speeding, yielding, gap acceptance, lane maintenance, visual scanning, signaling, and adjustment to stimuli errors, occur, for example, in drowsy drivers, drunk drivers, older driver, and distracted drivers. Because occupational therapists work with people representative of these groups, practice opportunities exist for general client education, and/or screening, assessment, and intervention.

Driver Rehabilitation

Driver rehabilitation entails a process of intervention intended to ameliorate impairments or deficits related to the driving task or alternative methods of mobility.[4] The outcome of driver rehabilitation is successful implementation of an intervention or course of action that facilitates safe mobility and transportation. Driver rehabilitation usually starts with screening, assessment, intervention, and discharge.

Screening

أول مرحلة بال Driver Rehab هي ال Screening

Screening is the process preceding an assessment and entails obtaining and reviewing data, personal, social, and medical history, to determine the need for assessment and can be done by the person themselves, a proxy rater, or an evaluator.[4]

الهدف كيف بعرف احتاج بتحتاج Further assessment or Not?

Self-Screening

For self-screening, an individual obtains and reviews his or her own data to determine the need for further assessment.[17–19] A limited number of valid and reliable screening tools for identifying at-risk older drivers exists.[20] For example, Roadwise Review, a self-screening, lacks predictive validity to detect at-risk drivers.[21] The DriveSafe and DriveAware self-screening accurately predicts the on-road performance of older drivers,[22] but as standardized in a country where driving occurs on the left side of the road. A drawback of self-screening tools is an overestimation of the client's driving abilities.[23]

(المبالغة في التقدير) اعلى من قدراتهم الحقيقية

ما بتربط هاي ال Self Screening

Proxy Screening

For proxy screening, an individual may obtain and review data to determine the need for assessment of another person.[19,24] The Fitness-to-Drive Screening Measure (FTDS),[24–30] a free, web-based screening tool with established reliability and concurrent criterion validity, identifies at-risk older drivers using a proxy rater that includes family members, close friends, or caregivers who have been passengers in the past 3 months and who are able to rate the driver's skills. The FTDS measure takes approximately 20 minutes to complete and generates one of three driver-risk classifications, indicating that *critical safety concerns* exist that must be addressed immediately; or *some safety concerns* exist with early signs of needing intervention; or *no safety concerns* are present and the driver can continue driving. Based on the specific driver-risk classification, the FTDS provides risk mitigation resources,

Further assessment ← يا سواقتك Safe ما بدها

Sign assessment ← يا سواقتك فيها شوية Concern

يا سواقتك ما بتربط ← وبده يوقف سواقة

including a listing of all **driver rehabilitation specialist (DRS)**, as well as recommendations indicating the next logical step to pursue. For more information on the FTDS, see http://fitnesstodrive.phhp.ufl.edu/us/.

Evaluator Screening

Evaluator screening is the process through which a professional, skilled in using a specific screening tool, obtains and reviews data to determine the need for assessment for a specific individual. For example, an occupational therapist may decide to use the Mini-Mental State Examination (MMSE) to screen for general cognitive functioning or the Optec 2500, a visual analyzer machine used for visual screening, to detect aspects of visual functioning, such as contrast sensitive or peripheral vision.[31] The driving literature supports the notion that an individual who scores lower than 24 on the MMSE should not be considered for an on-road assessment and that individuals who show impairment in visual fields or contrasts should be referred to an eye doctor prior to conducting a **comprehensive driving evaluation**.[31] These tasks are well within the scope of the occupational therapy generalist, who may, when using these methods, help with detection of potential client factors that may impair **fitness to drive**.

Driving Assessment

Assessments may include clinical assessments, simulated driving assessments, or on-road assessments.

Clinical Assessment

Occupational therapist generalists or (certified) DRSs use clinical assessments to quantify and better understand client factors pertaining to visual (e.g., visual acuity or contrast sensitivity), cognitive (e.g., executive functions or working memory), motor (e.g., range of motion or upper extremity coordination), and other sensory (e.g., proprioception or stereognosis) functions. All of these functions, to a greater or lesser extent, are important for the task of driving. Although it is beyond the scope of this chapter to provide a comprehensive discussion on evidence-based clinical assessments that may be correlated to, or predictive of, driving performance in medically at-risk populations through the life span, the reader is referred to the text on Driving and Community Mobility that contains an excellent chapter on this topic.[31]

Driving Simulator Assessment

The driving simulator, a clinical tool, with excellent face, content and construct validity, as well as reliability for real-world driving performance, is increasingly used in occupational therapy practices.[32] This technology uses a computer-controlled environment that presents selected aspects of the driving experience considered representational of real-world driving.[4] Clients may "drive" a scenario in which their accelerator, brake, and/or steering responses are being assessed. Usually, the data obtained from the assessment allow for objective measurements of the users' responses to designated driving tasks.[33] This method is particularly appropriate for clients who want to return to driving, but who may still pose considerable risk on the road, and who are, therefore, not ready for on-road assessments. Caution needs to be executed in making generalizations to real-world driving, as simulated driving is a representation of reality and not reality itself.

On-Road Assessment

The on-road driving assessment, sometimes referred to as the behind-the-wheel assessment, is conducted in real-world circumstances (i.e., driving a real vehicle, on the road, while encountering traffic in a dynamic and complex environment). The on-road assessment may incorporate a fixed-route or a variable-route design. The first is usually standardized, and elements of this course, such as the number of straights, and left and right turns, or the setting (residential area, city and highway driving), are controlled for, while the latter allows for options to assess a client's specific needs, such as an older driver who only wants to drive in daylight and in the absence of peak traffic hours to a selected setting in the community. This assessment, conducted by a **certified DRS**, is the criterion standard for evaluating a driver's fitness-to-drive abilities.[34–36] The assessment can be performed in closed road or open on-road conditions. The first may entail no-traffic such as a closed road circuit, or limited traffic such as experienced in a deserted parking lot. These venues allow the driver to become comfortable with the assessment vehicle and the vehicles controls and to perform basic maneuvers in preparation for on-road driving. Each condition has its own innate characteristics pertaining to setting, traffic density, maneuvers (e.g., left turn, stop, yield, straight driving), or

other road users that provide challenges for achieving a reliable and valid approach to determine a client's fitness to drive. Inability to control for these and other environmental conditions (i.e., time of the day, weather, roadway conditions, or other drivers' behaviors) may increase error during the evaluation.[37] As such, the final outcome of the on-road assessment (e.g., a pass/fail determination) may be prone to subjectivity. Table 31-1 illustrates selected on-road assessments and the percentage agreement between the driving maneuvers/behaviors and the driving environment. In a study by Odenheimer et al.,[35] the percentage representation of each on-road assessment against the components (driving maneuvers/behaviors and the driving environment) varies from 58% with minor representation of the environment to a 100% representation of the driving environment.[36]

Table 31-1. Selected Fixed Based Road Courses Matching Driving Maneuvers and Behaviors with Environmental Components[36]

		ENVIRONME						
	DEMOGRAPHICS	PARKING LOT	RESIDENTIAL AREA	SUB-URBAN AREA		CITY		HIGH
Driving maneuvers/behaviors	Course length and/or distance in time	Familiarize with test vehicle; basic maneuvers; angle/perpendicular parking; backing up	Low traffic/speed <35 km/hr (22 mph); R/L turn; adjust to stimuli; straight drive; school zones	Med. traffic with speed <70 km/hr (44 mph); changes, straight drive, traffic signals, pedestrian traffic	Four-lane road with med.-high traffic and speed >70 km/hr (44 mph)	Specific inter. include: yield, right of way, stop sign, four-way stop, two-way stop, traffic light protected/unprotected; traffic circle	Busy shopping center, pedestrians, grocery carts, other obstacles	Vehicl maneu such a parkin adjust speed backin up, stoppi
Source								
Michon[38]	NA	x	x	x	x	x	X	x
Odenheimer et al.[35]	16 km (10 miles) 45 minutes	X	**X**	–	–	x	X	x
Justiss et al.[34]	24 km (15 miles) 52 minutes	**X**	**X**	**X**	**X**	**X**	–	x
Korner-Bitensky et al.[39]	– km 45–60 minutes	X	x	x	–	x	x	x
UWO On-Road assessment[36]	36 km (23 miles) 60 minutes	**X**	**X**	**X**	**X**	**X**	**X**	**X**

R, right; L, left; Med., medium; inter., intersection; – km, no documented kilometers; **X**, driving maneuvers/behaviors and environment; x, driving maneuvers/behaviors no mention of environment; "with/major/minor/no environment" denotes the source mentioned all/major/minor/no environmental descriptions; % of components (driving maneuver, behavior, or environment) represented in each source; NA, not applicable; "–", information is not available; UWO, University of Western Ontario.

Naturalistic Driving Assessment

To overcome the deficits associated with the on-road assessment, naturalistic driving studies have emerged.[40–42] These studies are using a methodology to monitor or evaluate driving behavior via instruments or sensors installed in a driver's own vehicle. Such instruments provide an objective driver identification (ID), as well as driving data for each trip, and operate in a passive manner, requiring no interaction from the driver. Of course, collaboration with skilled team members is necessary to instrument the vehicle, activate the sensors, extract the data, and interpret the data. The main benefit of this method pertains to the objectivity inherent to the assessment process.

Comprehensive Driving Evaluation

This method entails a thorough assessment of an individual's personal, medical and driving history, driving knowledge, driving skills, and driving abilities by a DRS who may or may not be certified. The comprehensive driving evaluation generally includes clinical assessments of sensory/perceptual, cognitive, or psychomotor functional abilities; an on-road assessment; an outcome summary; and recommendations for an inclusive **community mobility** plan.[3,31] The outcome of the comprehensive driving evaluation results in a determination of whether the person is fit to continue to drive, requires rehabilitation prior to continued driving, or needs to consider driving cessation and alternative sources of community mobility. This information is usually related to the driver, who is accompanied by a caregiver, within the context of the multidisciplinary team. Clinicians need to be familiar with their state policies, as some states (e.g., California or Oregon) require mandatory reporting to the Department of Motor Vehicles, whereas others (e.g., Florida or Michigan) require voluntary reporting.

Interventions to Support Driving

Although evidence exists to support visual, cognitive, motor, and other sensory assessments in driver rehabilitation,[31] less is known about the efficacy and effectiveness of driving interventions across medically at-risk populations.[43,44] As such, occupational therapists are encouraged to design interventions based on best practices and sound clinical reasoning.[3] As is the case for other rehabilitation interventions, occupational therapists who are interested in driver rehabilitation make use of interventions pertaining to remediation, compensation, and adaptation.

- ***Remediation of driving skills:*** Drivers who show a lack of the necessary driving skills or a deterioration of driving skills may be referred to a DRS for behind-the-wheel training. In this case, the DRS may want to work on tactical skills (i.e., those maneuvers necessary for stopping, going, turning, yielding, overtaking, backing up, parking, or other general vehicle control maneuvers).
- ***Compensation:*** Imagine an older driver with depth perception challenges who stops too close to the lead vehicle at a traffic light. In this case, the occupational therapist may suggest improving the stopping maneuver, via teaching the client a compensatory method of "tire stopping." This means that the client needs to stop at a distance from the lead vehicle so that he or she can see the back tires of that vehicle. Such a compensatory method ensures safe vehicle positioning, while also overcoming the depth perception challenge.
- ***Adaptation:*** Much has been written on vehicle adaptations, and the reader is referred to the text on Driving and Community Mobility for an expansive overview of adaptive equipment used in the vehicle.[45] Such adaptations enable drivers to continue to manipulate the primary (i.e., steering, accelerator, and brake) and secondary (e.g., wind shield wipers, emergency brake, headlights) controls. Such adaptations are particularly helpful for clients with medical conditions impairing their function necessary to operate the vehicle and its controls. For example, orthopedic conditions, such as upper limb deficiencies and amputations; neurological conditions, such as hemiparesis, hemiplegia, or Parkinson disease; or spinal cord injuries, such as paraplegia, pose severe functional limitations on the client's abilities to safely operate a motor vehicle. Table 31-2 illustrates examples of diagnostic groups, the potential limitations in function, and the adaptive equipment necessary to overcome the functional deficit. Note that most all vehicle adaptations need to be implemented with great caution, requiring the clinical reasoning skills of a DRS and the technical installation skills of a vehicle modifier who meets the industry safety standards. Because this area of practice requires specialty knowledge, we are only introducing the occupational therapist to the concepts of vehicle adaptation and encourage referral of the client to a DRS for all vehicle modifications.

Table 31-2. Examples of Adaptive Equipment for the Vehicle Based on the Client's Diagnosis, Limitation in Function, and Function of the Adaptive Equipment to Overcome the Limitation

DIAGNOSIS	LIMITATION IN FUNCTION	ADAPTIVE EQUIPMENT	FUNCTION OF THE ADAPTIVE EQUIPMENT
Left upper limb deficiencies	Inability to use bilateral hands for turning maneuver	Spinner knob	Enables adequate steering wheel maneuvering with right hand
Right lower extremity amputation	Inability to manipulate gas and brake with the right lower extremity	Left foot accelerator and brake pedal	Enables client to manipulate accelerator and brake with the left lower extremity
Hemiparesis or paraplegia	Inability to use right lower extremity for accelerator and gas control	Push left-angled hand control or a push-rock hand control	Accelerator is controlled by pulling backward and brake is controlled by pushing forward on the hand control.

Driving Aids

- A driver who is physically disabled requires provisions for the following:
  - Ability to enter and exit the car
  - Ability to operate primary and secondary controls
  - Ability to secure himself/herself, along with any mobility equipment that cannot be separated from the driver's body

Steering Devices

سهل علم القيادة

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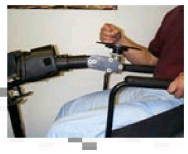
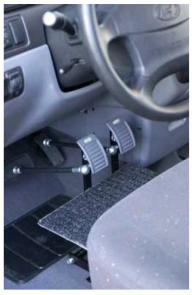


Vehicle modifications



Extended pedals

للبنزين والبريك

Left-foot gas Pedal

hemiplegic اذا عنده
السوا يتم الكبر
اليسار للتنين

Vehicle Adaptation

right hand
الفتاة

Turn-signal crossover

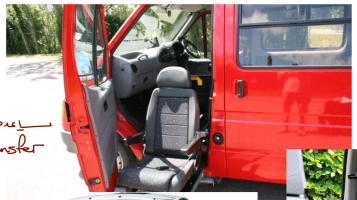
loss of one side of the body (left)

Vehicle Adaptations

البنزين والبريك
بواسطة
الايدين

Mechanical Hand control

they Can't Use their Upper ext



ساعده بال
transfer

كرسي
ينفتح
ال PWC

بال كرسي السائق
بدخل بال PWC وبضل
يوقف فيه

DIAGNOSIS	LIMITATION IN FUNCTION	ADAPTIVE EQUIPMENT	FUNCTION OF THE ADAPTIVE EQUIPMENT
Parkinson	Rigidity in the neck and trunk with difficulty in turning to perform a backing up maneuver	Wide angled mirrors or back-up camera to	Provides a rear view scene without the need to turn the trunk
Quadriplegia	Inability to be securely positioned due to limited trunk control	Torso strap for injuries at or above T-12	Torso strap secures the position of the client during driving and especially during turning maneuvers.

Community Mobility

Community mobility is defined as "moving around in the community and using public or private transportation, such as driving, walking, bicycling, or accessing and riding in buses, taxi cabs, or other transportation systems."[1] Access and availability of transportation is a critical facilitator for **community participation** and inclusion.[54–56] The National Council on Disabilities reported in 2015 that the lack of "accessible transportation represents one of the chief barriers to participating in economic and community life."[57] Research examining community participation defines this construct as "active involvement in activities that are intrinsically social and either occur outside of the home or are part of a nondomestic role."[2] Community participation can include integrated employment, health services, social events, education, and religious activities.[58] Recent research identified that perceived transportation and community mobility were predictors of participation restrictions in individuals with physical disabilities.[56] Older adults associated a sense of freedom and independence to participate in essential and desired activities with driving and community mobility.[59] As community participation is related to overall QoL,[60,61] self-determination,[60,61] and independence,[62] it is essential to identify needed supports and minimize barriers such as transportation to community participation. Transportation is integral to getting to many activities that are considered essential in the everyday lives of people, such as work, school, errands, medical appointments, and socialization with friends and family.

Types of transportation options for community mobility include personal, public, commercial, and supplemental transportation. Personal transportation involves moving around in the community using one's body and vehicle.[63] This includes traveling in a private automobile (i.e., being driven or driving), using other motorized or nonmotorized mobility devices (e.g., wheelchairs, scooters, biking), and walking. Driving oneself is often a preferred mode of transportation for many, although it may not be a viable option, because of newly acquired or progressing physical and/or sensory disabilities. Whereas driving is considered a privilege, community mobility using other forms of transportation is often considered a right. **Personal mobility devices** are used for community mobility when a person cannot walk or has limited walking ability. Public transportation options are available to everyone and involve moving more than one person at a time.[63] This includes both fixed-route transportation such as buses, trains, subways, and ferries, and **paratransit** services for individuals who have impairments that limit access to the fixed-route services. Supplemental transportation is an option for certain individuals, although availability can vary based on type of disability, age, and location. This type of transportation is often offered through volunteers from nonprofit or community-based services for individuals who are unable to use existing transportation services or need flexible travel options.[64] Finally, commercial transportation services are provided as for-profit entities most often paid by the individual. These can include taxis, transportation network companies (e.g., Uber/Lyft), shuttle/van services, and commercial carriers of airlines and trains. For shorter distances, commercial transportation options are now more affordable with the inception of transportation network companies and can provide greater flexibility than public transportation options. Although, for some of these options, there are accessibility issues for certain consumers when the transportation companies do not meet American with Disability Act (ADA) standards.

CASE STUDY: JULIO

To best illustrate the interplay of deficits in client factors, the assessment process, as well as the interventions necessary to address driving or community mobility, we introduce the reader to Julio via the following case study.[46,47]

Patient Information

Julio is a 15-year-old male, in the ninth grade, and living in the community with his parents and four siblings. He has never had a learner's permit or a driver's license. His parents are curious to know if he is ready to start driving. Currently, he is walking to school with his siblings, and although independent in way finding, he needs safety cues for scanning traffic before crossing the streets. Julio has a diagnosis of autism spectrum disorder (ASD) and shows characteristics consistent with attention-deficit hyperactivity disorder (ADHD). He has

received occupational and speech therapies for the past 6 years and is taking medications to manage the symptoms of his medical conditions. His parents reported no medication side or adverse effects.

Clinical Assessments

Julio completed clinical assessments, summarized in Table 31-3, to assess his visual, cognitive, and motor functions. The results indicated that he had intact visual acuity, peripheral vision, depth perception, color discrimination, and phorias. He had intact central vision processing speed, divided visual attention, and selective visual attention skills. His overall ability to integrate visual information with a specific motor response was mildly impaired. Julio had poor skills for simple sequencing, was mildly impaired with complex sequencing and had limitations in attention shifting and planning, and showed difficulty with gross motor coordination and strength.

Table 31-3. Julio's Clinical Tests, Description, and Scoring

CLINICAL ASSESSMENTS	DESCRIPTION AND SCORING	JULIO'S RESULTS
***/**Vision: Optec 2500 Visual analyzer (Stereo Optical, Inc.)[48]**		
Six tests on vision	Acuity, peripheral vision, phorias, depth perception, and color discrimination	Intact for all tests
***/**Visual Attention and Processing Speed[49]**		
UFOV risk index	Categories 1–5; 5 = very high risk for motor vehicle crashes	Intact: very low risk: category 1
Visual Motor Integration[50]		
Beery visual motor integration test[50]	30 possible points (30 = adequate age-based visual motor functioning)	Impaired at the 32nd percentile: 26/30 correct
Cognitive Assessments		
**Comprehensive trail making test[51]	Five trails tests measuring simple to complex sequencing	Overall impaired at the second percentile level
Symbol digit modalities test[52]	Measures motor, visual, speech, and executive functions and attention skills with the highest possible score of 110	Impaired: completed 39/110 symbols; 37/39 symbols were recorded with correct numbers
Motor Assessments		
Bruininks-Oseretsky test of motor proficiency (BOT-2): brief form[53]	Eight subtests of fine motor precision, motor integration, manual dexterity, bilateral coordination, balance, running speed and agility, upper limb coordination, and strength, with a total possible score of 88 points	Intact: percentile rank 38th Average score: 76 points
***Driving Simulator Assessment**		
****Simulator Operational Skills[a]**	Measured return demonstration of understanding and using vehicle controls on a VAS (0–10, where 10 = perfect demonstration)	Impaired: mild coordination deficits total VAS = 7.35/10
Driving Errors34		**Number of Driving Errors**
Lane maintenance errors: including wide turns or encroachments	The lateral lane position of the vehicle	30
Visual scanning errors	Visual scanning monitored at intersections and during lane changes	6
Lane position errors	The anterior and posterior position of the vehicle in relationship to the stop line	0
Speed error: over or under	Intact speed measured within 5 mph of the posted speed limit, or appropriate speed regulation required by maneuver (e.g., turn)	11 over speeding 0 under speeding
Signaling	Correct use of turn signal: timing, direction of signal, and appropriate use for the maneuver	0
Attention to stimuli	Correct response to changes in traffic light, posted signs, and traffic	2
Gap acceptance: overcautious or unsafe	Appropriate gap acceptance during the five unprotected left turns	2 over cautious 0 narrow
Divided attention (DA) responses	Five tasks of DA measured by the client honking when a diamond on the simulated scenario changed to a triangle	1
Speed of response to DA	Speed (norm < 11 seconds) at which the subject honks after the stimuli is presented	1.60 seconds
Number of collisions	Collisions involve either driving off road or hitting a vehicle or a pedestrian.	0

CLINICAL ASSESSMENTS	DESCRIPTION AND SCORING	JULIO'S RESULTS
Number of errors at intersections	All errors across the 11 intersections	Impaired at the 50th percentile. Errors: 52

*No adolescent norms available;

**percentile data based on the total driving errors of 18 neurotypical peers in the parent driving study.[46]

[a]The questionnaire measuring operational skills can be obtained from the primary author of this study.

UFOV, useful field of view; VAS, visual analog scale; WFLs, within functional limits.

Driving Simulator

Julio became confident and comfortable with the driving simulator via return demonstration of understanding and using the vehicle controls. Then, he drove a basic scenario with minimal traffic in an interactive driving simulator with a 180° field of view that afforded an immersive experience, as displayed in Figure 31-2. The DRS sat in the passenger seat and recorded his driving errors throughout the drive. Julio drifted beyond the limits of his lane multiple times on straight roads and showed mild impairments with motor coordination when turning the steering wheel. Although *lane maintenance* errors are common among novice drivers, the frequency of Julio's errors and lack of his awareness to correct for such errors suggested significant impairments related to visual motor integration and attention shift, which are critical abilities for the task of driving. Julio did not perform *visual scanning* for traffic at cross streets or during lane changes. He approached all intersections with excessive *speed,* causing him to make wide turns to the right and encroaching into oncoming traffic on the left. He was unable to learn from his errors. Julio drove through two red lights and reported that he did not think it was necessary to stop, reflecting his poor understanding of traffic flow as a cause for collisions, and showing *adjustment to stimuli errors*. He made two *gap acceptance errors* related to being overcautious, potentially owing to being a novice driver. Although he had severe difficulty with recognizing the *divided attention* symbols, his speed of response on the one divided attention task was good.

Figure 31-2 The University of Florida's STISIM M 500W interactive driving simulator with life-size images projected on the screen and simulator controls integrated with a 1997 Dodge Neon.

Occupational Therapy Recommendations

Based on the results of the clinical assessments and Julio's simulated driving performance, the Certified Driver Rehabilitation Specialist (CDRS) had to make a decision regarding Julio's readiness to drive. Relevant questions that the CDRS asked to decide if Julio was a candidate for driver training are:

- Was it appropriate to begin driver training?
- If driver training was appropriate, were there skill deficits that require occupational therapy intervention for remediation of underlying skills prior to driver's education?
- If no occupational therapy was required, would traditional driver's education be recommended?

- If driver training (with or without occupational therapy) was not recommended, is driving a possibility in the future?
- If driving was an appropriate future goal, what predriving skills needed to be further developed before learning how to drive?
- If driving was not a likely goal, what community mobility goals should be recommended?

Summary of the Short-Term Goals and Intervention

Based on Julio's personal and medical history, as well as the findings of the clinical assessments and simulated driving performance, the CDRS recommended that Julio was not ready to start driving. The short-term goal is to refer Julio to an occupational therapist to address a community mobility plan and the use of alternative transportation options.

In the next section, the reader will learn more about community mobility as a facilitator for participation.

The physical environment often serves as a facilitator or barrier for community mobility.[56,65] The ADA, established in 1990, included guidelines for public and private transportation for individuals with disabilities.[6] According to the ADA, public transportation systems are required to provide information on services in accessible formats for individuals with disabilities, as well as needed equipment and facilities (i.e., lifts, ramps, securement devices, signage, communication devices) that are in good operating condition.[57,66] Additionally, public transportation systems must (1) provide adequate time to board and exit a vehicle, (2) allow service animals, (3) provide designated seating and signs on fixed-route systems, and (4) ensure training for operators to provide safe operation of vehicles and equipment, as well as assistance for individuals with disabilities. Specific architectural requirements exist (e.g., lifts and ramps; stop announcements; illumination of stepwells and doorways; slip-resistant surfaces for door and steps) for fixed-route transportation systems to improve accessibility.[66] Architectural requirements are also required for transit facilities (e.g., accessible paths of travel, boarding ramps, bus shelters). Transit agencies must provide paratransit services, an origin to destination service, where fixed-route services exist for individuals with disabilities who qualify. Private transportation entities (e.g., airport/hotel shuttles, private buses, taxis) that provide services to the public are required by the ADA to meet accessibility requirements. If the vehicles are not accessible to an individual with a disability, the transportation entity must have an equivalent service that provides an accessible vehicle to travel to the same place at the same time and at the same cost. Taxis are not required to purchase accessible automobiles, although they cannot discriminate against individuals with disabilities by refusing service, use of assistant animals, stowing of mobility devices, or charging higher fares.

The ADA provides specific guidelines to promote environmental access to transportation. Environmental factors are some of the most significant barriers to community mobility for individuals with a variety of physical disabilities.[56,65] A fit between the unique characteristics of the person and the environment is essential for successful community mobility. Although there are many environmental barriers to community mobility that must be considered during the occupational therapy assessment and intervention process, in 2010, the Beverly Foundation identified five A's that were common barriers for the aging population and those with disabilities.[67] These included availability, accessibility, affordability, acceptability, and adaptability (see Table 31-4). All of these are important considerations when evaluating the fit between the environment and the person.

Table 31-4. Environmental Factors Impacting Transportation Usability[67]

ENVIRONMENTAL FACTOR	DESCRIPTION
Availability	Existence of transportation when needed
Accessibility	Transportation is reached and used in light of riders' abilities/disabilities.
Affordability	The costs are within the users means or reimbursable.
Acceptability	Meets standards of cleanliness, safety, courteous/helpful operators
Adaptability	Modification can be made for disabilities and special needs.

Assessment of Community Mobility

Although assessment and intervention to promote community mobility is not a new area of practice for occupational therapists working with individuals with physical disabilities, it continues to be an emerging area where roles and

needs have expanded significantly in recent years. This is likely due to the increased awareness of transportation as a primary barrier to community participation and engagement.[54,56] Occupational therapy assessments and interventions for community mobility are diverse and varied based on the types of transportation, service delivery model, and the unique needs of the individual with physical disabilities, although all have common elements (Fig. 31-3).

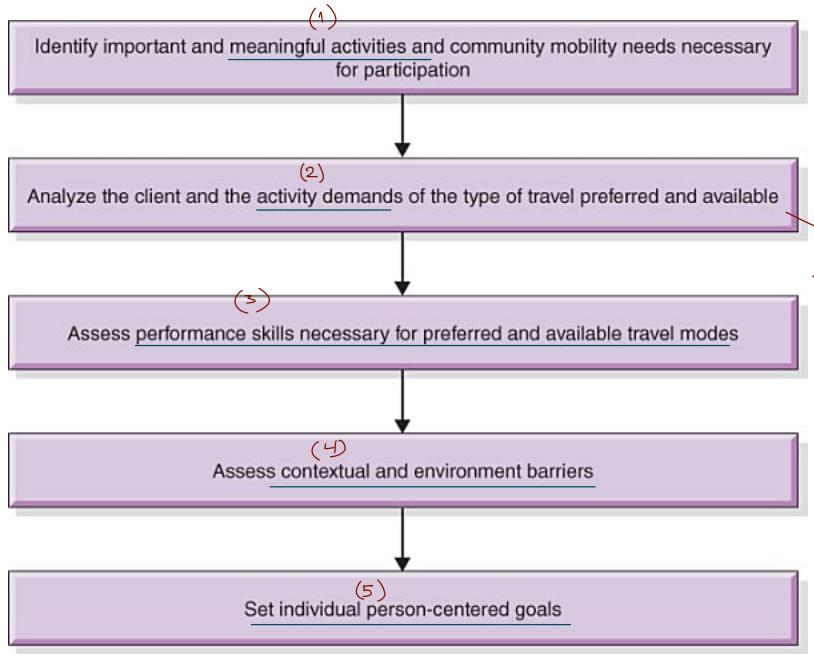


Figure 31-3 Assessing community mobility.

As in any occupational therapy evaluation, it is essential to obtain an occupational profile of the client to determine community mobility needs specific to the individual.[68] A person-centered approach in which the client is maximally engaged in the assessment and goal development process is essential. This includes identifying the person's important and meaningful activities and providing detailed information about their typical routines in order to understand the necessary mobility requirements and intervention for participation. Typical modes of transporting oneself or being transported by another person may change after an acquired physical disability or a change in a health-related condition making it necessary to reassess the way in which one travels in the community. This could include a change in the type of transportation mode (e.g., using paratransit instead of the taking the bus to get to a medical appointment) or the way in which the person travels in a previously used transportation mode (e.g., using hand devices to operate a car after sustaining an injury resulting in paraplegia).

Once community mobility needs are identified in the occupational profile, it is important to analyze the client and activity demands of the preferred or chosen types of travel modes. This will help determine individualize interventions based on the demands of the type(s) of travel and common environmental factors associated with specific types of travel. An example is an individual using a manual wheelchair due to a recent acquired physical injury who wants to use public transportation to visit his or her family or go shopping for essential items. After leaving home, he or she will need to be able to mobilize the wheelchair safely through the neighborhood to the bus stop and get on and off the bus. Environmental factors are an essential consideration when assessing the activity demands and can serve as either a facilitator or a barrier. For example, the presence of curb cuts on the pavements and a life with a wheelchair securement system on the bus could determine whether the use of a bus from the home is a safe and feasible option.

The evaluation process needs to consider both person factors (i.e., sensory issues, decreased balance, and motor deficits) and environmental barriers and facilitators to community mobility.[69,70] This allows the therapist and person to develop community mobility goals that match the strengths and needs of the person to environments that promote successful community mobility. The type of transportation often determines the necessary skills needed by the person. Although this can vary significantly, person factors[68] such as cognitive (i.e., attention, perception, memory, executive functions), sensory (i.e., vision, hearing, balance), and motor functions (i.e., endurance, control of

voluntary movement, gait patterns) are often necessary skills for successful community mobility outcomes. Tools available that assess specific aspects of community mobility includes the Functioning Everyday with a Wheelchair (FEW),[71] a criterion-referenced tool that assesses how wheelchair features help the client complete ADLs in situations such as outdoor mobility and use of personal/public transportation. The Life-Space Assessment documents where and how often clients travel and any assistance needed.[72] The Community Mobility Assessment evaluates physical and cognitive abilities of individuals with traumatic brain injuries to determined safe community mobility.[73] Finally, the Assessment of Readiness for Mobility Transition[74] provides a method to assess the emotional and attitudinal perceptions of aging adults who acquire a health condition that requires a transition from driving to other methods of transportation for community mobility. Many of these tools assess both person and environmental factors.

Environmental factors consist of both physical and social aspects of the environment that impact on community mobility.[58] The five A's (i.e., availability, acceptability, accessibility, adaptability, and affordability, see Table 31-4) identify a set of environmental factors that impact transportation usability in the senior and general disability population. This may provide helpful guidelines to assess common environmental factors that can serve as barriers or facilitators to transportation for individuals with disabilities. Observations of the person in the community environments and using preferred transportation is an important component of the assessment and intervention process. For occupational therapy provided in inpatient settings, including community mobility screenings in the assessment process can help prepare a person for discharge and transition back into the home and community environments. Additionally, occupational therapists can complete evaluations for public and private community-based organizations to identify barriers to safe and accessible transportation in these environments.

Interventions to Support Community Mobility

Occupational therapist's role in community mobility interventions varies, based on the unique needs of his or her clients and the service delivery model.[75–77] Interventions can be person-centered, such as developing specific foundation skills necessary for mobility (e.g., balance, motor planning, sensory awareness); providing travel training (e.g., pedestrian skills, training to use public transportation); providing assessment, education, and training to acquire and use mobility equipment and devices; and instruction in wheelchair skills and safe wheelchair transport. Occupational therapists may provide parent and caregiver education on how to transport family members and guardians safely. Occupational therapists can also provide organization-based interventions such as assessments for paratransit eligibility and consultation for ADA compliance with transportation providers and community-based organizations.

Community Mobility Plans (CMPs)[78] provide a conceptual framework to guide the intervention process (Fig. 31-4). The intended outcome of CMPs is to facilitate more independent transportation for community mobility necessary for participation in life activities deemed important by the individual. Person-centered interventions are integral to supporting self-directed goals of the individual that result in better community mobility and outcomes.[79,80] Community mobility plans outline a person-centered intervention in which the individual first identifies their community mobility goal(s) (e.g., traveling independently to work, coordinating transportation to a social outing with friends) and their preferred methods of transportation (e.g., bus, driving). A targeted assessment of the five A's, travel skills, and resource availability is completed to guide individualized intervention and supports. Using information from the assessment process, goals and preferred methods of transportation are reassessed and adjusted or modified as needed, based on the resource availability, individual's travel skills, and the ability to reduce barriers associated with the five A's. Individualized travel training and supports to obtain needed resources are provided with the targeted outcome of attaining their self-determined goals.

🚵 Case Example: Young Adult with Spinal Cord Injury

Client:

شاب عمره 28 سنة عنده إصابة في الحبل الشوكي (T12) ويستخدم كرسي متحرك يدوي. يريد العودة للجامعة والمشاركة في الأنشطة الاجتماعية.

🪜 Steps (مختصرة)

1. Identify meaningful activities:

الذهاب للجامعة، حضور المحاضرات، مقابلة الأصدقاء.

2. Analyze activity demands:

التنقل بالكرسي لمسافات طويلة، الصعود إلى المباني، المناورة بين الطلاب والممرات الضيقة.

3. Assess performance skills:

Summary

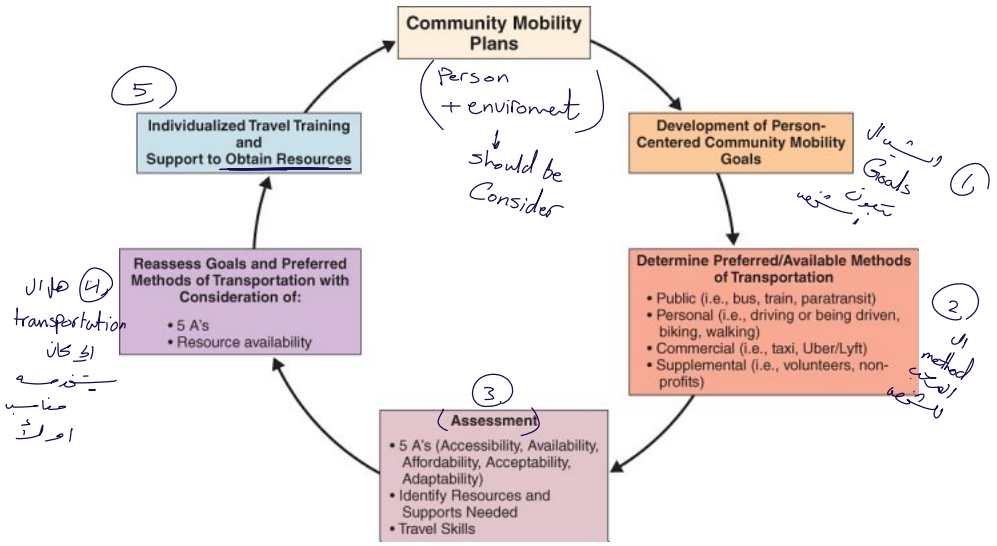


Figure 31-4 Community mobility plans.

Community mobility plans are founded on the premise that community mobility is attained when both the person and environment factors are considered when developing supports and interventions. This includes enhancing the skills necessary to use the preferred modes of transportation, while also addressing the environmental barriers in order to improve the person and environment fit. It is critical to support the ability and skill development of the individual to use available transportation options, but other environmental barriers must also be addressed as part of the assessment and intervention process.

CASE STUDY: TRISTEN

Patient Information

Tristen is a 19-year-old female who is graduating from high school. She was diagnosed with cerebral palsy at birth and uses an electric wheelchair for functional mobility. She is able to independently mobilize in her house and around the neighborhood using the wheelchair. She has functional use of both his hands for all activities of daily living and most fine motor tasks. Tristen excels academically in school and enjoys reading and watching movies. She has strong verbal communication skills. Tristen is planning on starting courses at the local community college and will be continuing her part-time job at a bookstore located in a nearby mall upon graduation. She identified transportation as a main concern upon her graduation in her most recent Individual Education Planning meeting. She currently uses the school bus to get to classes each morning and is dependent on her parents to drive her to work on the weekend. Her work hours will expand after graduation and include weekdays. Tristen wants to be able to travel independently to work and school when needed, as both of her parents work during the day. She identified that she would like to learn to use the bus because there is a bus stop near her house that drops her off at the mall and a few blocks from the community college. She has taken a public transportation bus with a family member in the past, but has never done this on her own.

Occupational Therapy Recommendations

The school occupational therapist who provides transitional services completed an evaluation to assess Tristen's community mobility. She was able to navigate safely to cross streets and maneuver on pavements that had curb cuts. She was not familiar with how to purchase bus passes and plan a trip using the bus (i.e., find schedules, locate routes specific to her travel destinations). Her local transit system has accessible buses that include a lift and securement system for wheelchairs. The bus operators assist patrons in wheelchairs to use a lift to get on and off the bus and ensure proper securement. There were some streets that did not have proper curb cuts, resulting in a need for Tristen to mobilize further to another street to get to her destination. During bad weather days (e.g., rain or snow), Tristen was not able to maneuver her wheel chair safely to get to bus stops.

Summary of Short-Term Goals, Intervention, and Progress

Based on the evaluation of Tristen's community mobility and her self-identified priorities, additional occupation therapy services were added to her IEP to address the following community mobility goals: (1) Tristen will be able to plan and implement a fixed-route trip to the mall using a public transit bus when provided with supervision and environmental adaptations successfully in four out of five trips. (2) Tristen will obtain conditional paratransit eligibility and schedule trips when needed to get to work or school with minimal assistance 90% of the time.

After exploring available transportation options and resources, Tristen and the occupational therapist identified bus transportation as the preferred method of travel to get to work and school. She also wanted to obtain conditional paratransit eligibility for days when she was not able to use the bus because of environmental barriers related to the weather (e.g., inability to mobilize in wheelchair safety when the pavement was wet or had snow on it). Additionally, she needed paratransit services to travel to other destinations that were not easily accessible with public transportation (e.g., medical appointments). Tristen worked with the occupational therapist and transition team at the school to identify the necessary resources and contacts to obtain eligibility for paratransit. This involved completing intake paperwork, obtaining medical documentation, and participating in an in-person assessment with staff at the transit agency. She participated in travel training with the occupational therapist once a week. The occupational therapist used an established travel training curriculum[81] that focused on use of public transportation and worked on foundational skills necessary to travel in the community (e.g., safe wheelchair navigation/securement, safety awareness). This involved direct instruction in the following six core areas, along with individual travel training: (1) Interacting with Individuals in the Community; (2) Signs and Traffic Symbols; (3) Pedestrian Safety and Awareness of Surroundings; (4) Preparing for Your Trip; (5) Travel Practice; and (6) Expect the Unexpected: Dealing with the "What Ifs."

Tristen made significant progress toward her goal after consistent intervention over a 2- to 3-month period of time. She was able to plan trips (e.g., identify the appropriate buses and the schedules; plan out enough time to get to the bus stop) to both the local community college and her part-time job at the mall using public transportation. She worked with the occupational therapist to identify routes to the bus stops that had curb cuts for accessibility purposes. Tristen was able to physically get on and off the bus but continued to work on recognizing when her stop was nearing so she could pull the cord that indicated when she needed to get off. She was demonstrating awareness of safety 90% of the time but continued to need supervision when there were unexpected changes (e.g., construction on a typical pedestrian route, unfamiliar types of interactions with people in the community). Tristen obtained eligibility for paratransit services. She worked with the therapist on how to schedule trips on paratransit and was able to implement a trip without supervision. By the end of the school year, Tristen was using fixed-route bus transportation to get to work at the mall and was able to independently get to the local community college. She started to explore other routes on the bus and other modes of transportation (e.g., Uber and Lyft) to get to the houses of family members and friends.

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